A3C and GAE

Remember, we want $\hat{A}_\pi(s,u) = \hat{Q}_\pi(s,u) - V_\phi(s)$.

A3C and GAE are ways of formulating $Q_\pi(s,u)$.

By definition, $Q_\pi(s,u) = E\left[r_0 + \gamma r_1 + \gamma^2 r_2 + \dots \mid s=s_0, u=u_0\right] = E\left[\sum_{t=0}^{T} \gamma^t r_t\right]$

This has high variance because it depends on the stochasticity of the entire rollout.

A3C uses n-step estimation for $Q_\pi(s,u)$.

$$\hat{Q}_\pi(s,u) = E\left[\sum_{t=0}^{n-1} \gamma^t r_t + \gamma^n V_\phi(s_n)\right]$$

so, $n=1 \Rightarrow \hat{Q}_\pi^{(1)}(s,u) = E\left[r_0 + \gamma V_\phi(s_1)\right]$

$n=2 \Rightarrow \hat{Q}_\pi^{(2)}(s,u) = E\left[r_0 + \gamma r_1 + \gamma^2 V_\phi(s_2)\right]$

$n=3 \Rightarrow \hat{Q}_\pi^{(3)}(s,u) = E\left[r_0 + \gamma r_1 + \gamma^2 r_2 + \gamma^3 V_\phi(s_3)\right]$

⋮ ⋮ ⋮

low n has low variance because you are taking current reward and average of everything else. High n has high variance because you are introducing the stochasticity of your specific rollout.

GAE takes the exponentially weighted average of the above equations.

$$\hat{Q}_\pi(s,u) = (1-\lambda)\hat{Q}_\pi^{(1)} + (1-\lambda)\lambda\hat{Q}_\pi^{(2)} + \lambda^2\hat{Q}_\pi^{(3)}$$

Ex. EWA of 3 horizons of $\hat{Q}$.

note that $(1-\lambda)+(1-\lambda)\lambda+\lambda^2 = 1-\lambda+\lambda-\lambda^2+\lambda^2 = 1$

Then, $\hat{A}_\pi^{GAE} = \hat{Q}_\pi(s,u) - V_\phi(s)$

$= (1-\lambda)\hat{Q}_\pi^{(1)} + (1-\lambda)\lambda\hat{Q}_\pi^{(2)} + \lambda^2\hat{Q}_\pi^{(3)} - \left((1-\lambda)V_\phi(s) + (1-\lambda)\lambda V_\phi(s) + \lambda^2 V_\phi(s)\right)$

$= (1-\lambda)\left(\hat{Q}_\pi^{(1)} - V_\phi(s)\right) + (1-\lambda)\lambda\left(\hat{Q}_\pi^{(2)} - V_\phi(s)\right) + \lambda^2\left(\hat{Q}_\pi^{(3)} - V_\phi(s)\right)$ Let $\hat{A}_\pi^{(i)} = \hat{Q}_\pi^{(i)} - V_\phi(s)$

$= (1-\lambda)\hat{A}_\pi^{(1)} + (1-\lambda)\lambda\hat{A}_\pi^{(2)} + \lambda^2\hat{A}_\pi^{(3)}$

TDerror

Let $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$

$= (1-\lambda)\left[r_0 + \gamma V_\phi(s_1) - V_\phi(s_0)\right] + (1-\lambda)\lambda\left[r_0 + \gamma r_1 + \gamma^2 V_\phi(s_2) - V_\phi(s_0)\right] + \lambda^2\left[r_0 + \gamma r_1 + \gamma^2 r_2 + \gamma^3 V_\phi(s_3) - V_\phi(s_0)\right]$

δ_0

$\delta_1 = r_1 + \gamma V_\phi(s_2) - V_\phi(s_1)$

$+\ \delta_0 = r_0 + \gamma V_\phi(s_1) - V_\phi(s_0)$

$\delta_0 + \gamma\delta_1 = r_0 + \gamma r_1 + \cancel{\gamma V_\phi(s_1)} - \cancel{\gamma V_\phi(s_1)} + \gamma^2 V_\phi(s_2) - V_\phi(s_0)$

$= r_0 + \gamma r_1 + \gamma^2 V_\phi(s_2) - V_\phi(s_2)$

$= \hat{A}_\pi^{(2)}$

How to calculate GAE?

We collect $(s_0, a_0, r_0) \rightarrow (s_1, a_1, r_1) \rightarrow (s_2, a_2, r_2) \rightarrow s_3$

evaluate $V_\phi(s_0)$, $V_\phi(s_1)$, $V_\phi(s_2)$, $V_\phi(s_3)$

We have TD errors

$\delta_0 = r_0 + \gamma V_\phi(s_1) - V_\phi(s_0)$

$\delta_1 = r_1 + \gamma V_\phi(s_2) - V_\phi(s_1)$

$\delta_2 = r_2 + \gamma V_\phi(s_3) - V_\phi(s_2)$

From previous page, we have $\hat{A}_\pi^{GAE} = (1-\lambda)\hat{A}_\pi^{(1)} + (1-\lambda)\lambda \hat{A}_\pi^{(2)} + \lambda^2 \hat{A}_\pi^{(3)}$

where $\hat{A}_\pi^{(t)} = \sum_{k=0}^{t-1} \gamma^k \delta_k$ where $\delta_k = r_k + \gamma V_\phi(s_{k+1}) - V_\phi(s_k)$

so, $\hat{A}_\pi^{GAE} = (1-\lambda)\delta_0 + (1-\lambda)\lambda(\delta_0 + \gamma\delta_1) + \lambda^2(\delta_0 + \gamma\delta_1 + \gamma^2\delta_2)$

$= (1-\lambda)\delta_0 + (1-\lambda)\lambda\delta_0 + \lambda^2\delta_0 + (1-\lambda)\lambda\gamma\delta_1 + \lambda^2\gamma\delta_1 + \lambda^2\gamma^2\delta_2$

$= (1-\lambda+(1-\lambda)\lambda+\lambda^2)\delta_0 + ((1-\lambda)\lambda+\lambda^2)\gamma\delta_1) + \lambda^2\gamma^2\delta_2$

$= \delta_0 + \lambda\gamma\delta_1 + \lambda^2\gamma^2\delta_2$

Generalized: $\hat{A}_t^{GAE} = \delta_t + \lambda\gamma\delta_{t+1} + (\lambda\gamma)^2\delta_{t+2} + \cdots$

$= \delta_t + \lambda\gamma[\underbrace{\delta_{t+1} + \lambda\gamma\delta_{t+2} + \cdots}_{\hat{A}_{t+1}^{GAE}}] \Rightarrow \hat{A}_t^{GAE} = \delta_t + \lambda\gamma\hat{A}_{t+1}^{GAE}$

Implementation:

advantage = 0

for $t: T-1 \rightarrow 0$

$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$

advantage $= \delta_t + \lambda\gamma$ advantage